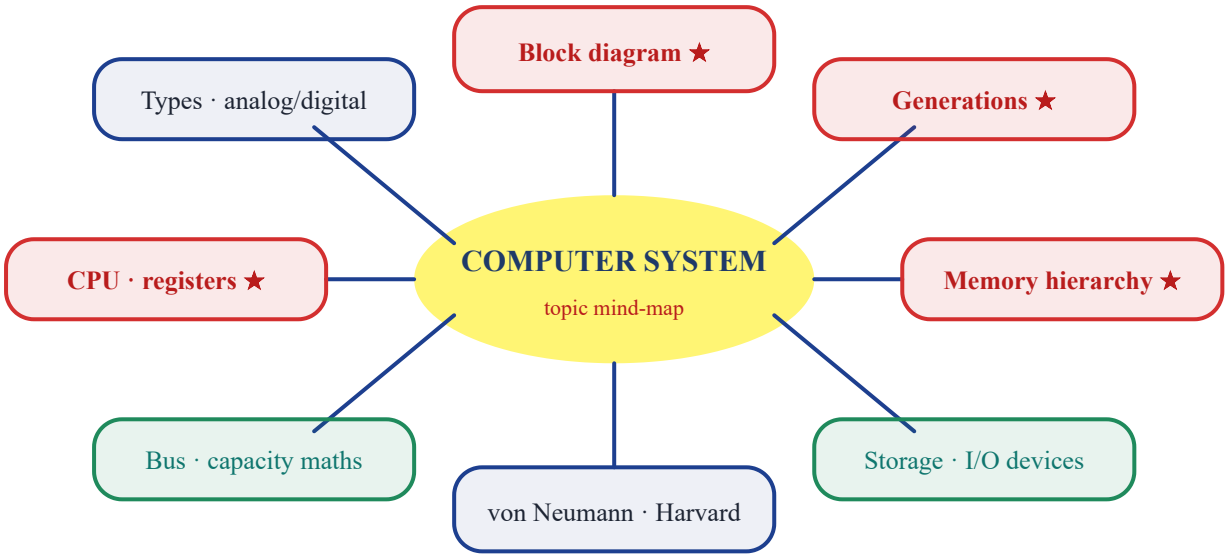
Computer Hardware & Architecture

Types · generations · block diagram · CPU · memory · storage · I/O: exam-ready in 11 sheets



↑ the whole lesson on one page: red = high-yield, green = applied / calculation

Syllabus coverage

"Introduction to Computer Systems: … **Computers**; **Major components of a computer**; **Hardware: processor, memory, I/O devices**; … **Basic architecture of a computer**". This notebook covers the hardware half. History of computing devices is lesson 1, software (OS, applications) lesson 3, the Internet lesson 4, number systems lesson 5; CPU design and the memory hierarchy in depth are in Chapter 7 (Computer Architecture).

High-yield list

① 5 functional units
② tube → transistor → IC → µP
③ PC, IR, MAR, MDR
④ registers > cache > RAM > disk
⑤ capacity = 2^n × word
⑥ OMR · OCR · MICR

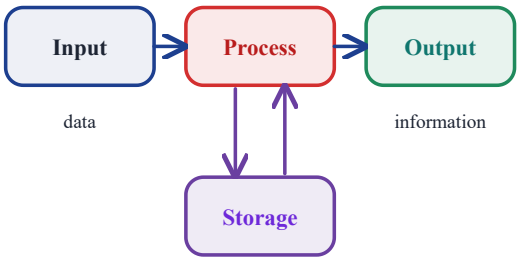
Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

✓ What a computer is; analog/digital … p. 2
✓ Types by size & power …… p. 3
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1 What Is a Computer?

Definition: A computer is a **programmable electronic machine** that accepts data (input), processes it under the control of a **stored program**, stores data and results, and produces information (output). Data = raw facts; **information** = processed, meaningful data.



the IPOS cycle: Input, Process, Output, Storage

Characteristic	Meaning
Speed	ns-level steps (MIPS, FLOPS)
Accuracy	errors are human (GIGO)
Diligence	no fatigue, works 24 × 7
Versatility	many kinds of tasks
Storage	huge, permanent, fast recall
Automation	runs a stored program alone

Speed units:

1 ms = 10^{-3} s · 1 µs = 10^{-6} s

1 ns = 10^{-9} s · 1 ps = 10^{-12} s

MIPS = million instructions / s

FLOPS = floating-point ops / s

Exam trap: Limitations: a computer has **no IQ**, no common sense and no feelings. **GIGO** = Garbage In, Garbage Out: wrong input or a wrong program gives wrong output, however "accurate" the machine.

2 Analog, Digital & Hybrid Computers

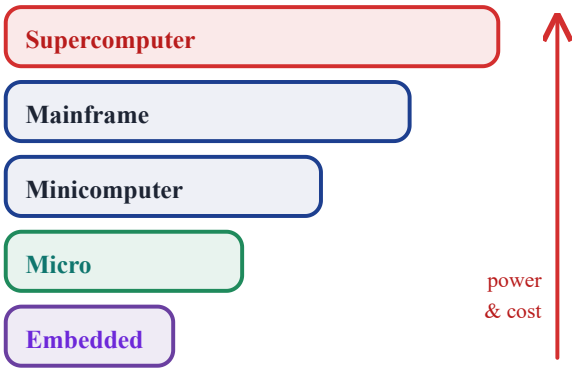
Type	Works on	Examples
Analog	**continuous** physical quantities (voltage, pressure, temperature); measures	speedometer, analog thermometer, voltmeter, differential analyser (Vannevar Bush, 1931)
Digital	**discrete** values, binary 0/1; counts; exact	PC, laptop, smartphone, calculator, digital watch
Hybrid	both: analog input → ADC → digital processing	**ICU patient monitor**, ECG machine, fuel pump, weather-forecast systems

Memory trick: **Analog measures, Digital counts, Hybrid does both.**

By purpose: **General-purpose**: any program (PC). **Special-purpose**: one fixed job (ATM, traffic-light controller, washing machine).

3 Types by Size, Power & Users

Class	Key feature	Examples	Typical use
Supercomputer	fastest; thousands of processors in parallel; speed in **FLOPS** (exascale = 10^{18})	CDC 6600 (first, 1964), Frontier, El Capitan	weather, climate, nuclear simulation
Mainframe	huge **throughput**; thousands of users at once	IBM Z (z16, z17)	banks, census, airline reservation
Minicomputer (midrange)	multi-user, tens to hundreds of terminals	DEC PDP-11, IBM AS/400	departments, mid-size firms
Microcomputer	built around a **microprocessor**; one user	desktop, laptop, tablet, smartphone	personal, office
Workstation	high-end micro: strong CPU/GPU, large RAM	CAD / graphics machines	engineering, design
Embedded	a **microcontroller** doing one job inside a product	washing machine, car ECU	control



power and cost rise upward; units sold the other way

Memory trick: Order by power: "**S**ome **M**ighty **M**achines **M**ake **E**rrors" = Super, Mainframe, Mini, Micro, Embedded.

Embedded systems are the **most numerous** computers: a car alone has dozens of ECUs.

> Seymour Cray = "father of supercomputing"

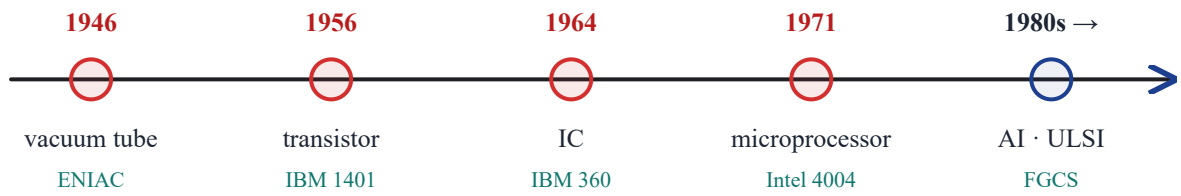
Exam trap: **Supercomputer** = raw **speed** on a few huge calculations (FLOPS). **Mainframe** = **throughput**: millions of small transactions from thousands of users. A bank's core transaction system → mainframe, not a supercomputer.

Server ≠ size: A server is a **role**: any computer that provides a service (web, file, database, mail) to client computers. A server can be a micro, a mini or a mainframe.

Recall drill ✍️: Micro family: desktop → laptop → tablet → smartphone → smartwatch (x86 or ARM chips). Supercomputers are ranked twice a year in the **TOP500**; Frontier (USA, 2022) was the first exascale machine.

4 Generations of Computers

Gen	Years (approx)	Switching device	Memory · software	Examples
1st	1946–1956	**vacuum tube**	drum, punched cards; machine code	ENIAC, EDVAC, UNIVAC I
2nd	1956–1963	**transistor**	magnetic core; assembly, FORTRAN, COBOL; batch	IBM 1401, IBM 1620
3rd	1964–1971	**IC** (SSI/MSI)	OS, multiprogramming, time-sharing	IBM System/360, PDP-11
4th	1971–present	**microprocessor** (LSI → VLSI)	semiconductor RAM; GUI, PCs, networks	Intel 4004, IBM PC
5th	present & beyond	ULSI, parallel processing	AI, natural language, robotics	Japan's FGCS project (1982)



the generation is named after its switching device

Invention dates:

Transistor 1947: Bardeen, Brattain, Shockley (Bell Labs)
IC 1958: Jack Kilby (TI); Robert Noyce 1959
Microprocessor 1971: Intel 4004, 4-bit, ≈2,300 transistors
ENIAC 1946: ≈18,000 tubes, ≈30 tonnes
UNIVAC I 1951: first US commercial computer

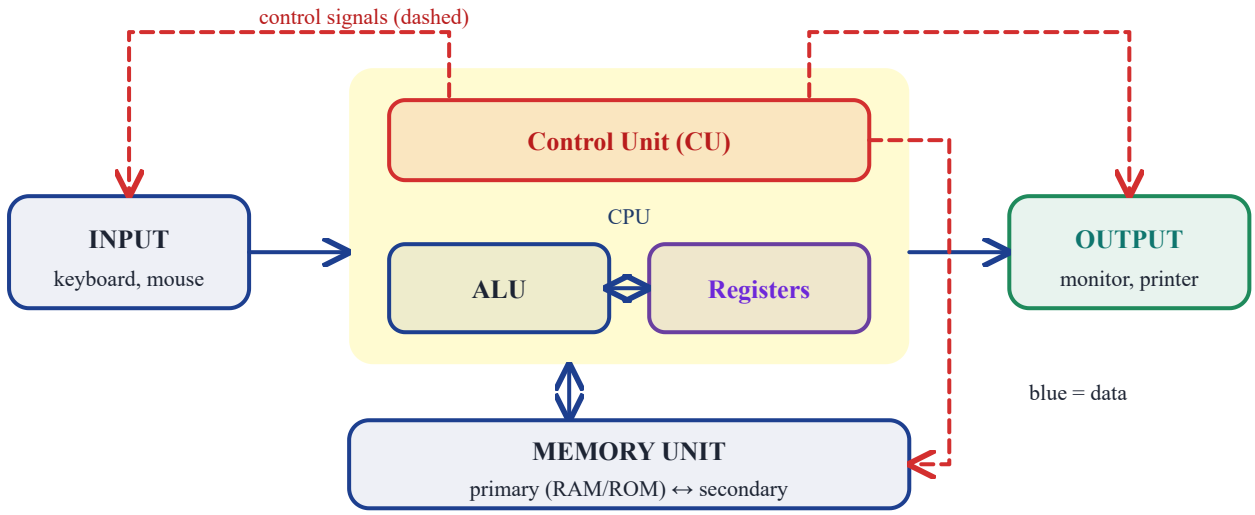
Scale	Gates per chip (approx)
SSI	< 10
MSI	10 – 100
LSI	100 – 10,000
VLSI	> 10,000
ULSI	> 1 million transistors

BD fact: the first computer in Bangladesh, an **IBM 1620** (second generation, transistor), was installed in 1964 at the Atomic Energy Centre, Dhaka.

Memory trick: "**V-T-I-M-A**": Valve (vacuum tube), Transistor, IC, Microprocessor, AI. Each step made computers smaller, cheaper, cooler and more reliable.

Exam trap: Invention year ≠ generation year: the transistor (1947) defines a generation that starts ≈1956, the IC (1958) one that starts ≈1964. Year ranges differ between books; the **technology** is asked. ENIAC was **not** a stored-program machine (EDSAC, 1949, was the first practical one).

5 Block Diagram: the Five Functional Units



input → CPU ↔ memory → output; the CU sends control signals to every unit

Unit	Function
Input	accepts data and instructions; converts them to binary
Memory	primary holds the running program and its data; secondary stores permanently
ALU	arithmetic (+ − × ÷) and logic (AND, OR, NOT, compare)
Control Unit	fetches and decodes instructions; issues control and timing signals to all units
Output	converts results from binary to human-usable form

CPU = ALU + CU + registers · whole system = CPU + memory + I/O, joined by the **system bus**.

6 System Bus & Stored-Program Architecture

Bus	Carries	Direction	Its width decides
Data	the data / instruction itself	**bidirectional**	bits moved per transfer
Address	location in memory or I/O port	**CPU → only**	n lines → 2^n locations
Control	RD, WR, interrupt, clock, reset	both (per line one-way)	which operations

> **Worked example ✍️:** Bus bandwidth = width × transfers per second: a 64-bit (8-byte) bus at 100 MHz moves $8 \times 100 \times 10^6$ = **800 MB/s**.

> **Stored-program concept:** Program instructions and data are kept in the **same** read-write memory, so a computer can be re-programmed by loading a new program. Described by **John von Neumann** in the 1945 "First Draft of a Report on the EDVAC".

von Neumann vs Harvard architecture

	von Neumann (Princeton)	Harvard
Memory	one memory for program + data	separate code and data memories
Buses	one shared set	two sets: code + data at once
Weakness	**von Neumann bottleneck**	more wiring, fixed split
Used in	PCs, servers (general purpose)	DSPs, microcontrollers (AVR, PIC)

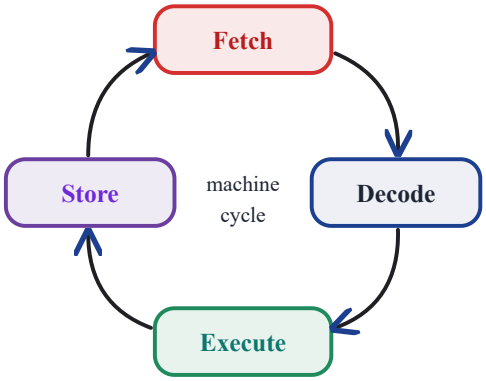
Modern CPUs are **modified Harvard** inside: split L1 instruction and data caches over one von Neumann main memory.

7 CPU: ALU, Control Unit & Registers

ALU: Does every calculation: add, subtract (multiply/divide by repeated add/shift), AND, OR, NOT, XOR, compare. Sets the **flags** (zero, carry, sign, overflow).

Control Unit: The "manager": fetches and decodes instructions and sends timing and control signals. It **does not process data**. Design: hardwired or microprogrammed (Ch 7).

Register	Holds
PC (Program Counter)	address of the **next** instruction
IR (Instruction Register)	the instruction being decoded / executed now
MAR (Memory Address Reg.)	address sent to memory (drives the address bus)
MDR / MBR (Memory Data/Buffer)	word just read from or about to be written to memory
AC (Accumulator)	one ALU operand and the result
Flag / Status, **SP**	condition bits · top of the stack



repeated for every instruction

1. **Fetch:** MAR ← PC; MDR ← M[MAR]; IR ← MDR; PC ← PC + 1
2. **Decode:** CU reads the opcode, finds operands
3. **Execute:** ALU operates, or data moves, or a jump loads PC
4. **Store** (write-back): result to a register or memory; check for interrupts

Memory trick: **"PC points, MAR asks, MDR carries, IR holds."**

Exam trap: PC holds the **address** of the **next** instruction, not the instruction. The CU never computes.

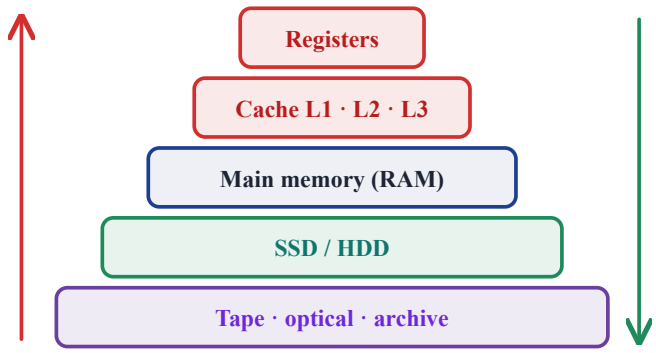
8 Clock Speed & CPU Performance

Clock period **T = 1 / f**: 1 GHz → 1 ns, **3 GHz → 0.33 ns**.

Word size = bits handled at once (32/64-bit). Multi-core = several cores on one chip.

Example ✍️: IC = 2×10^9, CPI = 1.5, f = 3 GHz
CPU time = IC × CPI / f = **1 s**
MIPS = f / (CPI × 10^6) = **2000**

9 Memory Hierarchy



red ↑ faster, costlier per bit · green ↓ bigger

Level	Access
Registers	< 1 ns
Cache	1 – 10 ns
RAM	50 – 100 ns
SSD	≈ 0.1 ms
HDD	5 – 10 ms
Tape	seconds +

Cache works because of **locality of reference**: temporal (the same item again soon) and spatial (neighbouring addresses next). Hit ratio h = hits / total accesses.

Worked example ✍️: Cache 10 ns, RAM 100 ns, h = 0.9 (simultaneous access): $T = h \cdot T_c + (1 - h) \cdot T_m = 9 + 10 =$ **19 ns**, five times faster than RAM alone. (Hierarchical access: miss costs 10 + 100 → T = 20 ns.)

10 Primary Memory: RAM & ROM

	SRAM	DRAM
Cell	flip-flop	capacitor
Refresh	no	**yes**
Speed	fast	slower
Cost/bit	high	low
Density	low	high
Used as	**cache**	**main RAM**

ROM type	Written / erased
Mask ROM	at the factory
PROM	once by the user (fuses)
EPROM	erase with **UV light**
EEPROM	electrically, byte by byte
Flash	electrically, in blocks

Memory trick: EPROM: **E**rase with **P**hotons (UV, quartz window). EEPROM: the extra E = **E**lectrically.

Exam trap: Primary ≠ volatile: **ROM is primary but non-volatile**. RAM, cache and registers are volatile.

11 Memory Units & Capacity Maths

Unit	Size
nibble · byte	4 bits · **8 bits**
1 KB	2^{10} = **1024** bytes
1 MB · 1 GB	2^{20} · 2^{30} bytes
1 TB · 1 PB	2^{40} · 2^{50} bytes
EB · ZB · YB	2^{60} · 2^{70} · 2^{80}

Memory trick: **K M G T P E Z Y**: "Kind Men Give Ten Percent Every Zakat Year"; each step × 1024.

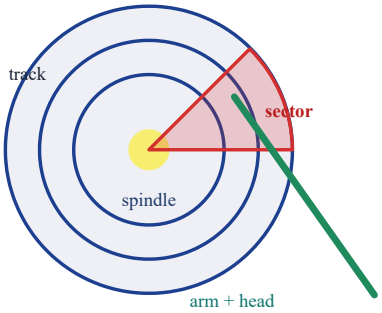
Exam trap: Disk makers count in 10^3 steps: a "1 TB" disk = 10^{12} B ≈ **931 GB** in Windows (2^{30}-byte GB, strictly GiB).

Key formula: n address lines → **2^n locations** · capacity = **2^n × word size** · m data lines = word of m bits

Worked examples ✍️:

1. 16 address lines, 8-bit word: 2^{16} × 1 B = 65,536 B = **64 KB**
2. 8086: 20 address lines → 2^{20} = **1 MB**; a 32-bit address bus → 2^{32} B = **4 GB**
3. A 1K × 8 chip has **10** address and **8** data lines. Chips for an 8K × 16 memory = (8K / 1K) × (16 / 8) = **16**

12 Secondary Storage



HDD platter

HDD: magnetic platters spinning at 5400 / 7200 rpm. Cylinder = the same track on every surface. Sector = 512 B (4 KB on new disks).

Access time = seek + rotational latency + transfer

SSD: NAND flash, no moving parts: faster, silent, shock-proof; limited write cycles. SATA or NVMe (PCIe, M.2).

Worked example ✍️: 7200 rpm → one turn = 60 / 7200 s = 8.33 ms → average rotational latency = half a turn = **4.17 ms**. Capacity of 8 surfaces × 1024 tracks × 64 sectors × 512 B = 2^{28} B = **256 MB**.

Medium	Capacity	Note
CD	650 / 700 MB	780 nm infrared laser
DVD	4.7 GB (1 layer), 8.5 GB (2)	650 nm red laser
Blu-ray	25 GB (1 layer), 50 GB (2)	405 nm blue-violet laser
Magnetic tape	TBs per cartridge (LTO)	**sequential access**, cheapest backup

Access: **random** (RAM) · **direct** (disk) · **sequential** (tape). Old floppy 3.5″ = **1.44 MB**.

13 Input Devices

Device	Reads / used for
Keyboard (QWERTY), mouse	text; pointing (mouse: Douglas Engelbart, 1960s)
Scanner	paper image → digital image
OMR Optical Mark Reader	filled bubbles: **MCQ answer sheets**, survey forms
OCR Optical Character Recognition	printed text → editable text
MICR Magnetic Ink Char. Recognition	**bank cheques** (E-13B magnetic-ink numbers)
Barcode / QR reader	product codes at POS, parcels
Light pen, digitiser, joystick	draw on screen; CAD drawing; games
Biometric reader, webcam, mic	fingerprint / iris (NID, attendance); video; sound

Memory trick: **OMR** reads **M**arks · **OCR** reads **C**haracters · **MICR** reads **M**agnetic ink on cheques.

14 Output Devices

Printer	Kind	Speed in
Dot-matrix	impact	cps
Line printer	impact	lpm
Inkjet	non-impact	ppm
Laser (toner, drum)	non-impact	ppm
Thermal (receipts)	non-impact	—

Monitor: CRT → LCD → LED-backlit LCD → OLED. Pixel = picture element; Full HD = 1920 × 1080.

Plotter: large engineering drawings (CAD). Also speaker, projector.

Both I/O: **touchscreen**, modem, NIC, multifunction printer.

Exam trap: Only **impact** printers can print carbon copies. Print quality is in **dpi**; screen resolution in **pixels**.

15 Ports, Motherboard & I/O Transfer

- **USB**: 2.0 = 480 Mbps, 3.0 = 5 Gbps, USB4 = 40 Gbps
- **HDMI** digital audio + video · **VGA** analog video (15-pin)
- **RJ-45** Ethernet · PS/2 old keyboard/mouse
- I/O transfer: programmed, interrupt-driven, **DMA** (block moves without the CPU)
- **Motherboard**: CPU socket, chipset, RAM (DIMM) slots, PCIe slots, SATA / M.2
- **BIOS / UEFI** firmware in flash: runs **POST**, then loads the OS
- **CMOS** + coin battery: clock and setup values
- SMPS converts AC mains to low DC voltages

Night-Before-the-Exam Sheet

Fact sheet:

IPOS: Input → Process → Output + Storage

Gen: tube → transistor → IC → µP → AI

Transistor 1947 · IC 1958 · µP 1971

Stored program: von Neumann, 1945

Address bus one-way · data bus two-way

T = 1/f · CPU time = IC × CPI / f

EPROM UV · EEPROM electrical

CD 700 MB · DVD 4.7 GB · BD 25 GB

CPU = ALU + CU + registers

ENIAC 1946 · UNIVAC I 1951

Intel 4004: 4-bit, first µP

Harvard = separate code/data memory

capacity = 2^n × word size

DRAM refresh · SRAM = cache

1 KB = 1024 B · 1 nibble = 4 bits

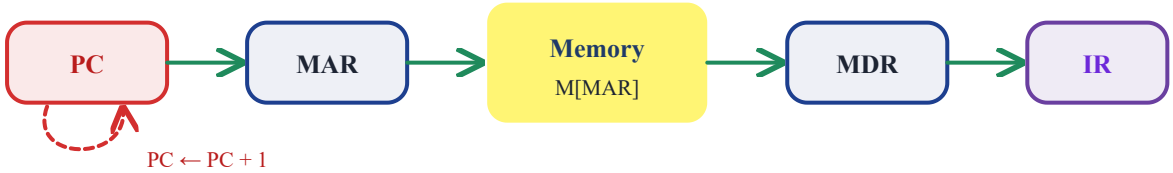
access = seek + latency + transfer

Top 12 one-liners for MCQs

1. Fastest memory: registers
2. PC holds next instruction's address
3. ALU does arithmetic and logic
4. RAM is volatile; ROM is not
5. Cheques: MICR · MCQ sheets: OMR
6. First computer in BD: IBM 1620 (1964)
7. Mainframe = throughput, many users
8. Supercomputer speed: FLOPS
9. ICU monitor = hybrid computer
10. Tape = sequential access
11. POST is run by BIOS/UEFI
12. DMA moves data without the CPU

Top traps: invention year ≠ generation year · ENIAC was not stored-program · primary ≠ volatile (ROM) · PC = address, IR = instruction · CU never computes · a "1 TB" disk shows ≈ 931 GB · supercomputer ≠ mainframe · server is a role, not a size

All tricks together: Analog measures, digital counts, hybrid does both · Some Mighty Machines Make Errors · V-T-I-M-A · PC points, MAR asks, MDR carries, IR holds · EPROM = Erase with Photons · Kind Men Give Ten Percent Every Zakat Year · OMR Marks, OCR Characters, MICR Magnetic



the fetch path in one line: PC → MAR → memory → MDR → IR

✓ Practice MCQs (exam pattern, 10 marks)

1. Which register holds the address of the next instruction?
(a) IR (b) MAR
(c) PC (d) AC

2. Which memory needs periodic refresh?
(a) SRAM (b) DRAM
(c) EPROM (d) Flash

3. Third-generation computers used:
(a) vacuum tubes (b) transistors
(c) microprocessors (d) ICs

4. 16 address lines, 8-bit word: capacity?
(a) 64 KB (b) 16 KB
(c) 128 KB (d) 32 KB

5. Banks read cheque numbers with:
(a) OMR (b) OCR
(c) MICR (d) BCR

6. Which bus is unidirectional?
(a) Data bus (b) Address bus
(c) Control bus (d) None

7. Which has the fastest access time?
(a) Cache (b) RAM
(c) Registers (d) SSD

8. An EPROM is erased by:
(a) UV light (b) electric pulse
(c) magnetic field (d) not erasable

9. The stored-program concept is credited to:
(a) Babbage (b) Turing
(c) Pascal (d) von Neumann

10. An ICU patient monitor is a … computer:
(a) digital (b) hybrid
(c) analog (d) mainframe

Answer key → 1-c · 2-b · 3-d · 4-a · 5-c · 6-b · 7-c · 8-a · 9-d · 10-b

Written question: Draw the block diagram of a basic computer system and state the function of each unit. What does the system bus do? 8 marks

1. **Diagram** (sheet 5): Input → CPU [CU + ALU + registers] ↔ Memory → Output, with dashed control lines from the CU to every unit. 2
2. **Input** converts data/programs to binary; **Output** gives results in human form; **Memory**: primary (RAM/ROM) holds the running program, secondary (HDD/SSD) stores permanently. 2
3. **ALU** does arithmetic and logic; **CU** fetches, decodes and issues control signals; **registers** hold operands and addresses. 2
4. **System bus**: address (one-way, picks the location), data (two-way, carries the word), control (RD, WR, interrupt, clock). 2

Before the exam ✓

- ✓ Can draw the 5-unit block diagram
- ✓ Know the 5 generations with devices
- ✓ Can name PC, IR, MAR, MDR, AC jobs
- ✓ Can order the memory hierarchy

Also practise (written):
① von Neumann vs Harvard
② SRAM vs DRAM; ROM types
③ 16K × 8 from 2K × 4 chips